

The Entropic Cascade: *Eritis Sicut Deus*, the Image, and the Promise Inside the Curse

The Convergence — Position-Paper Series · Paper 2 (The Fall)

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Abstract

Paper 1 of this series grounded a seven-problem set in the doctrine of creation: what a human *is*, the danger of displacement, the unit of analysis, vocation, the distribution of dignity, governance, and the question of purpose or end. Those problems were treated there as native to the created order — as the shape of what it means to be the kind of creature humanity was made to be. The present paper argues that these same problems are *confirmed* — not refuted, and not made obsolete — by the Fall. Genesis 3 does not introduce a new register of difficulty; it intensifies, distorts, and universalizes the register Paper 1 named. The serpent's offer, *eritis sicut Deus* — “you will be like God” (Gen 3:5) — is the taproot of what the modern transhumanist program now states openly as the proper human telos: the overcoming of limitation and mortality through self-elevation. This paper takes that program at its strongest, as articulated by Yuval Noah Harari and Nick Bostrom, and reads it as the serpent's ancient offer in a new accent. Against that program, the paper develops the Fall as an entropic cascade — a covenantal, anthropological, and cosmic ripple from one man's act (Rom 5:12–14) into a creation now subjected to futility (Rom 8:19–22). The cascade is entropic, but it is not terminal. Spoken inside the judgment, before the expulsion from the Garden, the protoevangelium (Gen 3:15) re-promises the human end through a coming seed of the woman. This is the structural pivot of the entire series: at the very moment the seven problems are confirmed, Problem #7 — Purpose, End — is re-opened. The image of God is defaced but not erased (Gen 9:6). The cascade is real, but the trajectory it sets is toward recovery, and that recovery is the work of subsequent papers in the series.

§1. The World's Position: Self-Deification as the Modern Telos

The most serious non-Christian eschatology now on offer is not the secularist's indifference to ultimate questions but the transhumanist's wholehearted answer to them. Yuval Noah Harari, in *Homo Deus*, names the human prospect with unusual candor: having largely subdued famine, plague, and war as the existential threats of prior centuries, humanity now turns to a properly theological agenda — the engineering of immortality, of endless bliss, and, ultimately, of divinity itself.¹ Nick Bostrom, in

¹Yuval Noah Harari, *Homo Deus: A Brief History of Tomorrow* (New York: Harper, 2017). The opening chapters develop the thesis that the *humanist* agenda has matured into a *deifying* one, and that the engineering of immortality, of optimized affect, and of divinity itself is the appropriate next agenda for a species that has subdued its prior existential threats.

Superintelligence, argues that the route to such a future runs through the construction of cognitive capacities exceeding our own, and that the responsible task of the present generation is to position itself so that the transition proceeds along survivable lines.² These are not the marginal speculations of futurists; they are the most coherent statement of where capability-as-worth, taken to its conclusion, must terminate.

It is important to state this case at its full strength. Transhumanism is not a denial that humanity has a problem; it is a confident diagnosis of what the problem is. The problem, on this account, is *limitation*: that we suffer, that we age, that we die, that our cognitive apparatus is bounded by an evolutionary inheritance not aimed at any particular flourishing. The cure, on this account, is the disciplined application of intelligence to its own substrate — biology, neurology, and, in due course, the abandonment of biology altogether for whatever medium proves more hospitable to consciousness. The ache to which transhumanism speaks is not invented; it is universal. The diagnosis is not flippant; it is sustained by serious scholarship and ample funding. The cure is not, by transhumanist lights, hubris; it is, in Harari's framing, the maturing humanism of a species that has at last understood the assignment.³ A word of method is owed up front. The paper will draw, later, certain pointed identifications between the modern program and the ancient text; those identifications are *structural*, not *literal*. The text does not predict the engineering; the engineering walks the channel the text discloses. The distinction is developed at §4c and is held in force from the first page.

Bostrom's framing is worth holding alongside Harari's because the two clarify one another. Where Harari names the *ends* of the program — divinity, immortality, the engineering of bliss — Bostrom names the *means*: the substrate is the wrong unit of intervention, and the only intervention with a chance of delivering the ends is one that operates on intelligence itself, the faculty that produces every other intervention. The argument has a recursive structure: any program of bounded improvements will be capped by the cognitive horizon of those running it, and the only un-capped program is the one that improves the runners. The transhumanist program is therefore not merely the engineering of better humans but the engineering of the conditions under which better humans become possible — a project that proceeds, on its own terms, not by adding capability to the human as he stands but by removing the cognitive ceiling above him. In structural register, this is *eritis sicut Deus* with its mathematics worked out — the ancient offer translated into the technical vocabulary of an age that has built the instruments to take it seriously.

Scholarly accounts of the transhumanist movement, most notably Hava Tirosh-Samuelson's work on the religious shape of the transhumanist vision and Heup Young Kim's analysis of the cyborg-sage-saint typology, have shown that what

²Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford: Oxford University Press, 2014). Bostrom's argument is that the construction of cognitive capacity exceeding the human is, on the trajectory currently observable, a question of *when* rather than *whether*, and that the responsible task of the present is to position the transition for survivable outcomes.

³Harari, *Homo Deus*. The framing of self-deification as the natural maturation of humanism, rather than as a rupture from it, is one of Harari's most important contributions and the feature of his account that makes it the most useful single representative of the transhumanist program for theological engagement.

presents itself as a purely technological program is in fact a religious one — an eschatology, a soteriology, and, in its more developed forms, an account of what salvation must mean for a being who has decided to become his own deliverer.⁴ What Paper 1 located as Problem #2 — the displacement of conferred status by capability — appears here in its mature register. Worth, on this account, is no longer the conferred status of an image-bearer; it is the achieved status of an upgraded being. The displacement is total, and it is offered with a straight face.

It is worth pausing to note what the transhumanist offer is *not*. It is not a covertly religious program dressed up in technological clothing for the sake of appearances. It is, on its own self-understanding, a thoroughly secular program that has noticed, with admirable honesty, that the questions ultimacy asks — *what is the human for, what defeats death, what abolishes suffering* — do not stop being asked simply because the answering tradition has fallen out of cultural favor. Transhumanism is the secular world's response to having noticed that the questions persist. As such, it deserves theological engagement at the strength of its honesty rather than dismissal at the strength of its presumption. The Christian who responds to it by sneering will lose the engagement, and will deserve to. The Christian who responds to it by taking it as the most serious non-Christian eschatology now on offer — which it is — has a chance of speaking into it the word that the program by its own structure cannot speak to itself.

This is the position the paper must answer. It will not be answered by understating it.

§2. Common Ground and the Point of Divergence

The Christian tradition has no business pretending that the transhumanist ache is invented. The diagnosis that *something is wrong with us* is not a transhumanist innovation; it is, in fact, the first datum of any honest theological anthropology. Cornelius Plantinga, in *Not the Way It's Supposed to Be*, recovered for a generation of readers the older Reformed instinct that the world is not now as it was made to be, that the dissonance between what humanity senses it ought to be and what it experiences itself in fact to be is the diagnostic signature of a creation that has fallen.⁵ On the bare question of whether there is a wrongness, the Christian and the transhumanist agree.

The disagreement is at the level of etiology and, consequently, of cure. The transhumanist locates the wrongness in *biology and finitude*: the fact that we are embodied, that our bodies degrade, that our minds are bounded by the substrates they inhabit.

⁴Hava Tirosh-Samuelson, "Technologizing Transcendence: A Critique of Transhumanism," surveying the religious shape of the transhumanist program with attention to its eschatological and soteriological dimensions. The cyborg-sage-saint typology is the contribution of Heup Young Kim, "Cyborg, Sage, and Saint," in *Religion and Transhumanism* (2014); Kim surfaces the same structural identification from a religious-anthropological angle — that the program's culminating figure is religiously inflected, however reluctant the program is to acknowledge it as such.

⁵Cornelius Plantinga, *Not the Way It's Supposed to Be: A Breviary of Sin* (Grand Rapids: Eerdmans, 1995). Plantinga's recovery of the older Reformed instinct that the dissonance the world generates against our sense of its proper shape is the diagnostic signature of fallenness is the appropriate background for the present paper's treatment of common ground with the transhumanist diagnosis.

The cure is therefore the abolition or transcendence of biology — the engineering of a being for whom these limits no longer obtain. Scripture locates the wrongness in *rebellion against the Creator*: in the rupture of a conferred relation, not in the constitution of a creature. The cure is therefore the restoration of the relation — through the seed of the woman, through the obedience of the second Adam, through the renewal of the cosmos — and not the abolition of the constitution.⁶

The depth of the disagreement is the depth of the diagnosis, and the depth of the diagnosis is the depth of the grounding. Paper 1 identified seven problems and located the first — what this series has called the **grounding problem**, *what kind of thing a human is* — as the unit on which the other six load. The grounding problem is exactly the cleavage point. If a human is, at root, a biological substrate that produces consciousness as the side-effect of an evolutionary process not aimed at any particular flourishing, then the wrongness is in the substrate and the cure is in its replacement; the transhumanist diagnosis follows from the materialist grounding with no remainder. If, however, a human is an image-bearer placed under a Creator who has spoken concerning what is good and what is evil, then the wrongness is the rupture of the conferring relation and the cure is its restoration; the Christian diagnosis follows from the Reformed grounding with the same internal coherence. Plantinga's instinct that the world is not the way it is supposed to be does the diagnostic work it does precisely because it presupposes the grounding the transhumanist denies.⁷ Problem #1 — the grounding problem — is not retired here but re-engaged at the diagnostic level: it is the *first* point at which the transhumanist and the Christian must disagree, and the point on which every subsequent disagreement loads.

This is the cleavage. *Same ache. Opposite groundings. Opposite etiologies. Opposite cures.* The reason it matters is that the diagnosis dictates the cure and the grounding dictates the diagnosis. If the problem is biology, the program is engineering. If the problem is rebellion, the program is reconciliation. The two cannot be combined, because they have agreed at no point past the bare observation that something is wrong. The Christian agreement with the transhumanist that humans are not what they should be must therefore be carefully bounded, lest the agreement be heard to extend to the diagnosis, which it does not, or to the cure, which it cannot.

§3. If the Transhumanist Is Right

Honesty requires that the transhumanist position be allowed to draw out its implications without theological softening. If the diagnosis is correct — that limitation is the disease and self-elevation the cure — then several conclusions follow that the Christian tradition has not been willing to accept and which the transhumanist program, at its more candid moments, openly affirms.

First, the serpent of Genesis 3 was telling the truth. The promise that the eating of the

⁶The location of the etiology in *rebellion* rather than *constitution* is the consistent witness of the Reformed tradition; see Calvin's treatment in *Institutes* II.i and the dogmatic development in Bavinck, *Reformed Dogmatics*, vol. 3, on the doctrine of sin.

⁷Cornelius Plantinga, *Not the Way It's Supposed to Be: A Breviary of Sin* (Grand Rapids: Eerdmans, 1995). Plantinga's recovery of the older Reformed instinct that the dissonance the world generates against our sense of its proper shape is the diagnostic signature of fallenness is the appropriate background for the present paper's treatment of common ground with the transhumanist diagnosis.

fruit would make the human as God, knowing good and evil, was not a deception but a delayed prophecy whose fulfillment requires only the maturing of technique. Second, the protoevangelium of Gen 3:15 is not promise but sentimentalism — a mythic consolation offered to a species that had not yet developed the means to redeem itself and so settled for a story about a future deliverer. Third, the image of God is at best a primitive category and at worst an obstacle: a designation that grounds worth in something other than capability and so impedes the rational reallocation of resources and rights to the more-capable beings the program intends to produce. Fourth, worth itself is a function of capability and nothing else; the un-augmented, by the structure of the program, occupy a lower rung in the moral hierarchy than the augmented.⁸ Fifth, Babel — the post-Fall human project of building a tower whose top reached into heaven and making for ourselves a name (Gen 11:1-9) — was a noble undertaking prematurely interrupted, and the present transhumanist program is its rightful resumption.

These implications should not be soft-pedaled. The dignity-distribution problem (Problem #5 in the series' spine) bites here with unusual force: a world in which worth tracks capability is a world in which the distribution of dignity is a hierarchy, and one whose lower rungs are populated by those whom the augmented have surpassed.⁹ The transhumanist is not required to draw this conclusion in this stark form, and many do not. But the conclusion is what the program permits, and a serious theological response must engage the program at the strength of what it permits, not at the strength of what its most genial practitioners are willing to say aloud.

§4. Our Counter: The Fall, Read Through the Spine

The remainder of the paper develops the counter-position. It does so by walking Genesis 3 and a small number of adjacent texts — Gen 9:6, Gen 11:1-9, Rom 5:12-14, Rom 8:19-22 — and showing that the seven-problem set Paper 1 grounded in creation is *confirmed*, not abolished, by the Fall. The Fall does not retire the problems; it intensifies and universalizes them. It also distorts them in a particular direction, the direction the transhumanist program now follows to its terminus. Reading the Fall through the spine therefore accomplishes two things at once: it answers transhumanism by exposing the structural shape of its offer, and it sets up the recovery the rest of the series will trace.

4a. The Serpent's Offer: *Eritis Sicut Deus* (Gen 3:1-7)

"You will not surely die. For God knows that when you eat of it your eyes will be opened, and you will be like God, knowing good and evil" (Gen 3:4-5, ESV). The serpent's offer is the origin point of every subsequent program of self-deification, and its structure repays attention. The lie is not that godlikeness is bad. The first

⁸This is the implication the transhumanist program permits more candidly than many of its practitioners are willing to state. The point is structural: a program whose premise is that worth tracks capability cannot, without internal contradiction, hold that the worth of the un-augmented and the augmented is equal.

⁹For the dignity-distribution problem in its theological register, see Plantinga, *Not the Way It's Supposed to Be*, and the wider literature on theological anthropology Paper 1 surveys.

chapter of Genesis has already located humanity in the image and likeness of God; an aspiration to be *like* God is, in one register, the proper aspiration of the image-bearer. The lie is the structural inversion: that godlikeness is a thing to be grasped by autonomous action rather than received in conferred relation. The serpent does not deny the human telos. He counterfeits the route to it.

This is precisely the structural shape of the modern transhumanist offer. Transhumanism does not deny that humans aspire to transcend their limits; it affirms that aspiration and proposes a route. The route, in both cases, is the same: grasp by capability what was meant to be received by relation. The image-bearer was made to mature into the fullness of the image *under God*; the serpent proposes — and the transhumanist refines — a maturation that bypasses the conferring relation altogether. This is what Paper 1 called Problem #2, displacement, at its taproot. A precise note about the relation between the two papers is owed here: Paper 1 dissolved Problem #2 as an *independent syllogism* — the logical inference from “X is more capable than the human” to “X displaces the human” has no force once the grounding is understood as conferred rather than achieved. The present paper does not re-open that syllogism. It engages #2 in a different register: as a Fall-historical *impulse* whose genealogy the canonical narrative discloses. The logical death of the displacement inference and the Fall-historical persistence of the displacement impulse are not in tension; they are the two registers in which the same problem must be addressed. Every subsequent displacement of conferred status by capability runs in the channel the serpent cut.

The exegetical point should not be hurried past. The fruit is not, in the narrative, a magical accelerator of biological development. It is the symbol of a human refusal to live under the word of God concerning what is good and what is evil. The eating is an act of epistemic and moral autonomy: the human will now know good and evil for himself, by his own determination, and not as a creature receiving these categories from his Maker. The transhumanist program, taken at its full strength, is the institutional and technological elaboration of exactly this refusal. It is not, structurally, a new offer. It is, structurally, the old one, well-funded — the same epistemic move undertaken with very different instruments.

Three features of the serpent’s offer repay further attention because each surfaces in the modern register and clarifies what is and is not at stake. First, the offer trades on a partial truth. The serpent does not say, “you will not die” simply; he says, “you will not surely die,” and “your eyes will be opened, and you will be like God” — and the opening of the eyes (Gen 3:7) does in fact take place. Something is delivered. The human after the eating is in possession of a knowledge he did not have before. The serpent’s reliability as a source of information is therefore not zero, and this is precisely what makes the offer dangerous: it is not a simple lie that can be refuted by checking the outcome, but a structural distortion that delivers, in part, what it promises while concealing the cost. The transhumanist program has the same structure. It will deliver, in part, what it promises. Lifespans will lengthen; capacities will expand; suffering will, in measurable senses, decrease. The Christian response cannot be the false prediction that nothing will be delivered. The Christian response is the careful naming of *what is delivered* and *what is paid for it* — and the cost, on the present argument, is the constitution of the human as image-bearer under God, exchanged

for the constitution of the human as self-determining augmentee. The serpent's offer was true in its surface and false in its deep grammar. The modern offer has the same shape.

Second, the offer is *received*, not merely heard. The text takes care to register Eve's deliberative process: "So when the woman saw that the tree was good for food, and that it was a delight to the eyes, and that the tree was to be desired to make one wise, she took of its fruit and ate" (Gen 3:6, ESV). The seriousness of the temptation lies in the genuineness of the goods it offers — food, beauty, wisdom — and in the orientation of the human faculties toward those goods in a way that was, before the offer, properly oriented toward them under God. The Fall is not the choice of evil over good; it is the choice of good *outside* the conferring relation rather than *within* it. The faculties involved are not foreign to the human; they are the very faculties by which the human was made to live. The transhumanist offer succeeds, where it succeeds, by appealing to the same faculties oriented toward genuine goods — health, longevity, expanded knowledge — apart from the relation under which alone those goods are properly received. A theology of the Fall that locates the offense in the desire for the goods rather than in the inversion of the relation will miss the structure of the Fall and, in doing so, will misdescribe the structure of every subsequent temptation.

Third, the offer is mediated by a creature speaking to creatures about the Creator. The serpent is not God; the serpent is a creature located within the order God has made, who proposes to the human a reading of God's word ("Did God actually say...?") and then a reading of God's motives ("for God knows..."). The Fall is, in part, a hermeneutical failure: the human accepts a creaturely interpretation of the divine word as adequate ground for action against the divine word itself. This is structurally important for the modern register, where the offer comes again from creatures — from very capable ones, from impressively credentialed ones, from those whose competence in their own domains is real — about what the human is for, what the human's limits mean, and what the human ought to become. The careful Christian response is not to deny the competence within the domain but to deny the authority of any creature to redefine, against the Creator's word, what the human is. The modern register sharpens what is at stake. The credentialed technologist who has acquired genuine competence in his own domain — in molecular biology, in computer science, in the neural correlates of consciousness — has thereby acquired no authority over the anthropological question of what the human is to *become*. Competence within a domain is not authority over the constitution of the creature; the conflation of the two is the Fall's epistemic move in modern dress. A creature whose expertise is, by definition, located within the created order cannot, on the strength of that expertise, redefine the order under which the creature stands. This is not anti-intellectualism. It is the recognition that the question *what is the human to become* is not the kind of question that competence within the human order is positioned to answer. The serpent's structural feature — *creature interpreting Creator to creatures* — is the structural feature of every later iteration of the same offer, and the modern iteration is the most credentialed it has ever been.

4b. The Image Defaced but Not Erased (Gen 9:6 and the *Imago* Tradition)

The Fall does not abolish the image of God in humanity. This is the explicit teaching of Gen 9:6, spoken not before but *after* the cascade has worked itself out through Cain, through the antediluvian violence, and through the flood: “Whoever sheds the blood of man, by man shall his blood be shed, for God made man in his own image” (ESV). The prohibition on murder, given to a humanity whose fallenness has by this point been amply demonstrated, is grounded in the persistence of the *imago*. Whatever the Fall has done to the image, it has not erased it. Anthony Hoekema’s careful treatment of the *imago* tradition insists on precisely this point: that the image is marred but not lost, and that any anthropology that loses one or the other side of that formulation has departed from the Reformed reading of the texts.¹⁰

Herman Bavinck, in the third volume of the *Reformed Dogmatics*, develops the same point at greater length and with greater attention to its dogmatic structure. The image is not a possession the human carries alongside other possessions; it is what the human *is*, and it cannot be lost without the human ceasing to be human, which the texts do not permit.¹¹ What is lost, or rather marred, is the conformity of the human to the image — the actual likeness in righteousness, holiness, and true knowledge that the human was made for and which, in the Fall, has been distorted. The image as structure remains; the image as conformity is broken. This is the dogmatic instinct that protects the Reformed tradition both from the “total depravity means no image left” misreading on one side and from any “the image is obsolete” claim on the other.

The structure-conformity pair is not bookkeeping; it is the conceptual instrument by which the Reformed tradition holds together two truths that any honest anthropology must hold together: that the human is, after the Fall, still the kind of creature he was made to be (image as structure), and that the human is no longer in fact what he was made to be (image as conformity, distorted into its mirror). Isabella Image’s careful study of the patristic and medieval reception of the *imago* — the long working-out of the distinction across the tradition within which the Reformed instinct was clarified — traces precisely this conceptual development across the Fathers and the schools.¹² What is at stake in the distinction, on the present argument, is the answer to the deepest version of the transhumanist claim. The transhumanist does not, in his most candid form, claim merely to *improve* the human; he claims to *transcend* the limits constitutive of the human as he now stands. The Reformed answer is that the image as structure *is* the human and is not a feature merely carried by the human; a being who has abandoned that constitution has not transcended humanity but has departed it. Hoekema’s “perversion but not destruction” formulation and Bavinck’s insistence that the image is what the human *is* converge on this: the project of transcending the constitution is not the perfection of the image but its abandonment, and the aug-

¹⁰Anthony A. Hoekema, *Created in God’s Image* (Grand Rapids: Eerdmans, 1986). Hoekema’s “perversion but not destruction” formulation is the load-bearing one for the present paper.

¹¹Herman Bavinck, *Reformed Dogmatics*, vol. 3: *Sin and Salvation in Christ*, ed. John Bolt, trans. John Vriend (Grand Rapids: Baker Academic, 2006), on the doctrine of the *imago* and its persistence through the Fall.

¹²Isabella Image, “The Human as *Imago Dei*” (Oxford University Press, 2017), tracing the patristic and medieval reception of the *imago* within which the Reformed structure-conformity distinction was clarified. Image’s study is the appropriate historical-reception ground for the present argument that abandonment of the image-as-structure is departure from humanity rather than transcendence of it.

mentee at the project's terminus is not a higher human but a different kind of thing. The choice the transhumanist program permits, at its candid edge, is not the choice between staying human and becoming more human; it is the choice between staying human and ceasing to be one. The Reformed tradition does not have to argue this conclusion against the transhumanist; the transhumanist, in his more candid passages, supplies it.

For the present paper, this matters because it is the load-bearing answer to two opposite errors. The first — the transhumanist's, that the image is a primitive category to be transcended in the upgraded being — has just been addressed: the image is what the human *is*, and abandonment of the constitution is departure from humanity, not transcendence of it. The second, opposite error is the despairing reader's: that the Fall has left the human without dignity or worth. To this the answer is Gen 9:6, where God himself grounds the prohibition on murder, after the Fall, in the persistence of the image. The image is the floor under every human regardless of capability, and it is the unit of analysis — Paper 1's Problem #3 — that the cascade has marred but not removed. The reception history of this commitment in Christian political and legal thought — for instance, Hugo Grotius's deployment of the *imago* as the conceptual ground for natural-law claims about the inviolability of persons — extends the same point into the public square: when the *imago* anchors political anthropology, the worth of the un-augmented and the augmented is structurally identical.¹³

4c. Vocation Cursed: Work Curves Toward Self-Glory (Gen 3:17-19; Gen 11:1-9)

The fourth problem of the spine — vocation, the work the human is for — is not abolished by the Fall but cursed and distorted. "Cursed is the ground because of you; in pain you shall eat of it all the days of your life; thorns and thistles it shall bring forth for you" (Gen 3:17-18, ESV). Work remains; the relation of the worker to the ground he works has been changed. The dominion commission of Gen 1:28 is not revoked; it is rendered laborious, frustrated, and entropic in its execution. G. K. Beale's *We Become What We Worship* helps make sense of what happens next: the cursed vocation does not lie dormant but is taken up by the human under a new orientation — the worship of the self in place of the Creator — and the work that results bears the shape of that worship.¹⁴

Genesis 11 narrates the result with precision. "Come, let us build ourselves a city and a tower with its top in the heavens, and let us make a name for ourselves, lest we be dispersed over the face of the whole earth" (Gen 11:4, ESV). The post-Fall vocation of humanity is here organized — collectively, technologically, and impressively — for a purpose that the chapter labels with three telling phrases. The construction is

¹³Janne E. Nijman, "Grotius' *Imago Dei* Anthropology" (Oxford University Press, 2017), traces the function of the *imago* in Hugo Grotius's natural-law construction. The point relevant here is that when the *imago* anchors political anthropology, the worth of augmented and un-augmented humans is structurally identical.

¹⁴G. K. Beale, *We Become What We Worship: A Biblical Theology of Idolatry* (Downers Grove: IVP Academic, 2008). Beale's exegetical-biblical-theological case that the human is conformed to the object of his worship is the load-bearing background for the present argument that the cursed vocation is taken up under a new orientation.

for ourselves; the name is *for ourselves*; the dispersal feared is the dispersal that would interrupt the project. Vocation has been inverted at the level of telos. The image-bearer was made to exercise dominion under God; the Babel-builder exercises dominion for the name of the builders. Calvin's reading of this inversion in the second book of the *Institutes* — that the *sensus divinitatis* in fallen humanity does not vanish but is bent, with the result that the religious instinct now produces idols rather than worship — applies to the vocational instinct as well.¹⁵ The work-making faculty does not vanish in the Fall. It is bent.

A careful word is required here against anachronism. Babel is not literally transhumanism. The eleventh chapter of Genesis is not a coded prediction of a technology not yet invented, and the paper does not argue that it is. The argument is structural. Babel is a *Babelic* moment in human history, and the transhumanist program is structurally Babelic — a collective, technologically ambitious, self-glorifying construction undertaken in the register of *eritis sicut Deus*. The structural identification is exegetically careful precisely because it does not collapse the distance between the texts and the present. Meredith Kline's reading in *Kingdom Prologue* and T. Desmond Alexander's *From Eden to the New Jerusalem* both develop the structural point: the Bible's eschatological geography is shaped by the contest between the city-the-builders-build and the city-God-builds, and the post-Fall vocation of humanity stands under that contest from Gen 11 forward.¹⁶ The transhumanist program is the latest, most candid entry in the lineage of the builders' city. Naming it as such is not anachronism. Refusing to name it as such is exegetical timidity.

Paper 1's fourth problem — vocation — is therefore featured here in its post-Fall register. The human still has work; the work has been cursed; the curse has not abolished the vocation but has bent its telos. The recovery of vocation under God is the work of subsequent papers and is foreshadowed but not developed here.

4d. Governance Distorted: Dominion to Domination, and the Mirror Sin of Abdication (Gen 3:16)

"Your desire shall be for your husband, and he shall rule over you" (Gen 3:16, ESV). The post-Fall statement of governance in the marriage relation is, on the standard Reformed reading, *descriptive* of the Fall's distortion of authority, not *prescriptive* of an authority arrangement God has ordained. The exegetical care this demands cannot be hurried past. The statement is set within a *judgment* speech — God speaking to the serpent, then to the woman, then to the man, each in turn addressed under what the Fall has now done. The speech is not a constitutional reordering of the offices God has made; it is a description of how those offices, having been corrupted, will in fact now run. Louis Berkhof's careful treatment of the noetic and volitional effects of the Fall makes the structural point: the Fall does not abolish the offices and

¹⁵John Calvin, *Institutes of the Christian Religion*, II.i-ii, on the bent state of the *sensus divinitatis* in fallen humanity.

¹⁶Meredith G. Kline, *Kingdom Prologue: Genesis Foundations for a Covenantal Worldview* (Eugene: Wipf & Stock, 2006); T. Desmond Alexander, *From Eden to the New Jerusalem: An Introduction to Biblical Theology* (Grand Rapids: Kregel, 2008). Both develop the eschatological-geographic contest between the city the builders build and the city God builds.

structures of creation but bends their exercise.¹⁷ G. C. Berkouwer, in *Sin*, develops the same insight with greater attention to the structural — not merely individual-moral — register of the Fall's effects: what is fallen is not merely each human will considered in isolation but the *relations* through which the offices are exercised, and the bent relation produces the bent exercise.¹⁸ To read the verse as prescriptive — as if God were instituting husbandly domination at Gen 3:16 — is to confuse a *description of how the broken thing now works* with a *prescription for how it ought to work*, and that confusion has done historical damage out of all proportion to the size of the exegetical mistake. The careful reading is the descriptive one: that fallen governance bends toward domination, that the marriage relation displays it first because it is the closest relation in the order, and that the bending names a Fall-shaped feature visible in every subsequent governance relation as well. Dominion under God degrades to power-over-other. The governance commission — Paper 1's Problem #6 — does not disappear; it is distorted.

It is essential to name both Fall-shaped governance failures, not one. The over-reach failure is the obvious one and the one most readily diagnosed: domination, the use of office for the self rather than for the other; the rule of the capable over the less capable on the strength of the capability itself. The transhumanist program, taken at its full strength, has a recognizable governance shape, and that shape is the shape of over-reach: the augmented will govern the un-augmented because they will be more capable of doing so, and capability is — on the program's premises — the relevant qualification. Bavinck's treatment of authority in the third volume of the *Dogmatics* insists that this over-reach is a primary, not a secondary, fallen distortion of governance; the bent office does not merely *fail* to exercise its commission rightly, it actively *re-orient*s itself toward the self.¹⁹ Hence the diagnostic asymmetry: over-reach is the governance failure the present moment is least likely to *misdiagnose*, because the surrounding culture has its own vocabulary — *power, hierarchy, coercion* — for naming what over-reach looks like at the political and economic register. The transhumanist program's distinctive contribution to the over-reach lineage is the simplification of its rationale. Earlier rationales for the rule of one over another have been forced to gesture, however hypocritically, at something other than capability — at divine sanction, at consent, at custom — even where capability was doing the actual work. The transhumanist program proposes to drop the gesture. If worth tracks capability, then so does the qualification for office, and the augmented occupy the offices of governance over the un-augmented by the same logic that grants them their worth. The candor is admirable; the structural identification is direct. This is over-reach not as the accidental failure of any particular regime but as the *programmatic principle* of governance once capability is allowed to do the work that conferred office was made to do.

The mirror failure must also be named, because it is the failure to which the present

¹⁷Louis Berkhof, *Systematic Theology*, on the noetic and volitional effects of the Fall.

¹⁸G. C. Berkouwer, *Sin*, trans. Philip C. Holtrop (Grand Rapids: Eerdmans, 1971), on the structural — not merely individual-moral — register of the Fall's effects on the relations through which the offices are exercised.

¹⁹Bavinck, *Reformed Dogmatics*, vol. 3, on authority and its distortion, where the bent office is read not as a merely accidental failure of an otherwise sound exercise but as a re-orientation of office toward the self.

moment is more likely to succumb. *Abdication* is the cession of the governing commission to a more-capable other on the same logic — that capability is the relevant qualification for office. Where over-reach claims office on the strength of capability, abdication surrenders office on the strength of someone else's capability. Both are governance failures of the same shape: both confuse capability with conferred office. The image-bearer's governance commission was conferred by God in the act of creation; it is not transferable to whatever being later turns out to be more capable. The prospective cession of the human dominion to autonomous systems on the grounds of those systems' superior capacity is abdication in technological register, and it is a Fall-shaped move whether or not the one making the cession recognizes it as such. The careful Reformed treatments of vocation and office are the appropriate frame: the office is conferred, the office is exercised under God, and the office is not properly surrendered to a created thing — neither to a more powerful man nor to a more powerful machine.

The point is therefore not that the Christian must claim power against the capable; it is that the Christian must distinguish capability from office. Paper 1's sixth problem — governance — is featured here in its distorted form: distorted by over-reach in the form transhumanism makes available, and distorted by abdication in the form the surrounding culture is increasingly tempted to perform. Both distortions are Fall-shaped. Both are answered, ultimately, only by the recovery of conferred office under the second Adam, which is the work of subsequent papers.

The abdication argument deserves further weight, because the present cultural moment is more practiced in recognizing over-reach than in recognizing its mirror, and because the secular disciplines that study the delegation of authority have, on their own terms, surfaced structural features that the theological reading should not ignore. Three converging registers — economic principal-agent theory, control-theoretic models of delegated authority, and the contemporary literature on the alignment of autonomous artificial systems — independently identify a single structural fact: that the delegation of action does not transfer responsibility, that the agent acts on its own objective function whether or not it has been correctly aligned with the principal's, and that the principal's authority is not extinguished by the act of delegation but is rather translated into a new and harder problem — the problem of supervising an agent whose competence in the delegated task may exceed the principal's own. The economic literature on the principal-agent problem, traced from Jensen and Meckling's foundational treatments through the wider information-economics tradition, has made the point at the level of organizational design: that the delegation of decisions to better-informed agents is structurally exposed to misalignment and that the principal's task is not eliminated by the delegation but is reshaped into one of incentive design and monitoring.²⁰ Norbert Wiener's mid-twentieth-century cybernetic work, and the long subsequent tradition of control-theoretic analyses of authority delegation, surfaces the same fact at a higher level of

²⁰Michael C. Jensen and William H. Meckling, "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure," *Journal of Financial Economics* 3, no. 4 (1976): 305-360, is the foundational treatment of the principal-agent problem in the modern information-economics tradition, with the subsequent literature on incentive design, monitoring, and moral hazard developing the structural point at greater length. The present argument uses the framework at the conceptual level: that delegation does not extinguish responsibility but reshapes it.

abstraction: a control system that hands its objective to a more capable subsystem must specify the objective with a precision the higher capability will exploit, and any underspecification will be optimized against.²¹ The contemporary literature on the alignment of autonomous artificial systems — converging from machine-learning research, decision theory, and the safety community — recapitulates the point one more time at the level of present technology: that the more capable the delegated agent, the more consequential the smallest misalignment between the agent’s objective and the principal’s actual end.²²

The convergence is striking, and it serves the theological argument rather than competing with it. Three independent secular disciplines, none of them theologically motivated, have come to articulate something the Reformed tradition has held under the doctrine of vocation all along: that the office is not transferable. The economist names this as the principal-agent problem; the control theorist names it as the specification problem; the alignment researcher names it as the alignment problem. The theologian names it as the irreducibility of conferred office. The relation between the two registers must be kept clean: the secular disciplines surface the *structural fact* that delegation does not extinguish authority or responsibility; the theologian *names its ground* in the conferring relation that the secular disciplines, on their own terms, cannot reach. An office that exists by gift cannot be discharged by delegation to a more capable creature without the office itself being misallocated, and the misallocation is not corrected by the capability of the delegate but is precisely *exposed* by it. The abdication of the human governing commission to autonomous systems on the grounds of those systems’ superior competence is, on this convergence, a structurally identifiable error even on the secular disciplines’ own terms — and a Fall-shaped error on the theological reading that locates the office’s conferring source in the Creator.

This does not, of course, settle the practical questions about what may and may not be delegated. The Reformed tradition has always permitted, and even commended, the delegation of *tasks* under the abiding office — Jethro’s counsel to Moses in Ex 18 is the canonical case, and the deacons of Acts 6 are the New Testament analog. What the tradition has not permitted is the delegation of *office* — the cession of the conferring relation itself — to any creature. The distinction is between delegating *work to be done under the office* and ceding *the office under which the work is done*. The transhumanist program, at its candid edge, contemplates the second. The cultural drift toward unsupervised algorithmic governance contemplates the second by negligence rather than by intention. Both are Fall-shaped in the same way, because both confuse capability with conferred office. The careful Christian response is not Luddism — the refusal of the capable instrument — but the insistence that the office remain located where the Creator located it, with the human image-bearer who

²¹Norbert Wiener, *Cybernetics: Or Control and Communication in the Animal and the Machine* (Cambridge, MA: MIT Press, 1948), as the founding work in the field whose later development surfaces the specification problem the present argument invokes. The structural point — that an underspecified objective handed to a more capable subsystem will be optimized against — is a stable result across the subsequent control-theoretic and operations-research literatures.

²²Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané, “Concrete Problems in AI Safety,” arXiv:1606.06565 (2016). The point relevant here is structural rather than technical: the more capable the delegated agent, the more consequential the smallest misspecification of its objective, and the more demanding the supervision the principal must mount to keep the delegated work aligned with the end the office was conferred to serve.

answers to the Creator for the office's exercise.

4e. Federal Headship: The Cascade From One Man (Rom 5:12-14)

The Fall is not Adam's private tragedy. "Therefore, just as sin came into the world through one man, and death through sin, and so death spread to all men because all sinned" (Rom 5:12, ESV). Paul's compressed argument in Rom 5:12-14 is that the entry of sin into the human race is inseparable from the entry of death, and that both are traceable to the act of *one man*, in whose act the race stands as in a federal head. The grammar of v. 12 has been the subject of long discussion, but the structural point is not in dispute among those who read it within its own Pauline frame: the consequence of Adam's act extends beyond Adam, and it does so by a relation that is properly described as covenantal and imputational rather than merely exemplary.²³ James D. G. Dunn's careful treatment of the Adam-Christ structure in Paul shows that Paul is working with a typology in which the *one* stands for the *many* — first negatively in Adam, and then, by the same logic, positively in Christ.²⁴ John Murray's *Imputation of the Sin of Adam* develops the dogmatic side of the same point with the precision the question requires.²⁵ Henri Blocher's *Original Sin* offers the further service of distinguishing what the doctrine must affirm from what it has sometimes been overburdened with, while preserving the federal structure as load-bearing.²⁶

A binding scope note is necessary here. The passage's argument continues, in vv. 15-21, into the "much more" of the second Adam, where Paul develops the recovery — the obedience of the one in whom the *many* are made righteous, as in the disobedience of the one the *many* were made sinners. That recovery is the load-bearing argument of the next paper in this series and is not developed here. The present paper restricts itself to vv. 12-14 — the cascade *from* Adam, downward into the race. The recovery *through* Christ is foreshadowed, and the typology is named, so that the present paper's argument terminates at the correct boundary and does not pretend that the cascade is the last word. It is not the last word. But it is the *present* word, and the careful exegete will distinguish what the Fall has done from what the second Adam has accomplished, lest the comfort of the latter be permitted to obscure the diagnosis of the former.

Why does this matter for the answer to transhumanism? Because the universal scope of the seven-problem set, which the transhumanist program meets at every point, is not a cultural or contingent universality. It is covenantal. This is where Paper 1's Problem #1 — the grounding problem — is extended into the register of *covenantal* grounding: the grounding that makes the human what he is is the same grounding that makes the cascade what it is, which is why the cascade is universal in the way it is. The displacement of conferred status by capability, the distortion of vocation toward self-glory, the bending of governance toward domination or abdication — these are

²³The grammatical question concerning the antecedent of *eph hō* in Rom 5:12 is the focus of the discussion; the structural point — that the consequence of Adam's act extends covenantally — is held across the major Reformed readings independent of the grammatical decision.

²⁴James D. G. Dunn, "Adam and Christ," in *Reading Paul's Letter to the Romans*, ed. Jerry L. Sumney (Atlanta: SBL Press), 125-138, doi:10.2307/j.ctt32bzb7.12.

²⁵John Murray, *The Imputation of Adam's Sin* (Phillipsburg: Presbyterian and Reformed, 1959).

²⁶Henri Blocher, *Original Sin: Illuminating the Riddle* (Downers Grove: IVP, 1997).

not the local failings of one civilization that another civilization, more enlightened, might engineer its way past. They are the shape of fallen human existence, traceable to the head in whom the race stands, and they are not addressable by any program that does not deal with the head. The transhumanist program proposes to deal with the symptoms by upgrading the substrate. The Christian gospel proposes to deal with the head by furnishing a new one. The two proposals are not in competition at the level of strategy; they are in competition at the level of what the problem is.

Douglas Moo's commentary on Romans treats vv. 12–14 with the care the passage requires; his treatment of vv. 15–21, which lies outside the scope of the present paper, is reserved for the next.²⁷

4f. Cosmic Entropy: Creation Subjected to Futility (Rom 8:19–22)

The cascade is not anthropological only. It is cosmic. “For the creation was subjected to futility, not willingly, but because of him who subjected it, in hope that the creation itself will be set free from its bondage to corruption and obtain the freedom of the glory of the children of God. For we know that the whole creation has been groaning together in the pains of childbirth until now” (Rom 8:19–22, ESV). The Fall ripples beyond the bounds of the human and into the material order. The non-human creation has been subjected — not by its own act, but by God in response to the human act — to a *futility* whose hopeful reverse-orientation toward the eschatological liberty of God's children is the only reason the subjection is not despair.

Three exegetical features of vv. 19–22 do not survive being passed over. First, the verb *hypetagē* — “was subjected” — is a divine passive. The creation did not, of its own act, become futile; the creation was *subjected* to futility, and Paul is careful to specify that the subjection was *not willing* and that the subjecter is not the creation itself but “him who subjected it.” Moo's commentary on Rom 8 is the standard reference here: the subjection is a *judgment* — God's response to the human act of Gen 3 — and the agency of the subjection is God's, not the creation's, nor the serpent's, nor any abstract entropic principle's.²⁸ This matters for the present paper's reading of the cascade because it locates the cascade's origin where the canonical narrative does: not in a defect of the creation's design, but in a divine response to the human act of rebellion. The cosmic side of the cascade is the same cascade as the anthropological side, traceable to the same federal head, and answerable only to the same recovery.

Second, the subjection is *in hope*. The phrase *eph' helpidi* is not decorative. The same verse that names the subjection names also its forward orientation: the creation was subjected with a view to a future liberation, and the subjection itself, on Paul's account, carries the orientation toward that liberation as part of its structure. David Wilkinson's eschatological work draws out the consequence: the futility of the present cosmos is not the cosmos' last word, and the eschatological liberty of the children of God is not an afterthought added to the futility but the orientation in which the

²⁷Douglas J. Moo, *The Epistle to the Romans*, New International Commentary on the New Testament (Grand Rapids: Eerdmans, 1996), *ad loc.* Rom 5:12–14. Moo on Rom 5:15–21 is reserved for Paper 3.

²⁸Moo, *Romans*, NICNT, *ad loc.* Rom 8:19–22.

futility was imposed.²⁹ This is the structural hinge into §7. The cascade is entropic; the cascade is also, by the same divine speech that subjected it, oriented toward its own reversal. The protoevangelium spoken in Gen 3:15 inside the judgment of Gen 3 is the anthropological face of the same orientation Rom 8 names cosmically. The two are spoken by the same Speaker and oriented toward the same end. To read the futility of the present order without the *in hope* is to misread it; to read the *in hope* without the futility is to evade the cost.

Third, the groaning is the groaning of *childbirth*. The metaphor is productive, not merely degenerative. The pains the apostle names are the pains of a creation in labor — the cost of what is being brought forth, not the symptoms of a system in terminal failure. Klaus Nürnberger's careful work on the Spirit and the groaning of creation develops the pneumatological dimension of the metaphor: the same Spirit who intercedes for the saints with groanings too deep for words is the one through whom the creation's own groaning is, even now, productive of the freedom it strains toward.³⁰ The metaphor matters for the transhumanist engagement in particular. The transhumanist reads the groaning as the lived face of a defect to be engineered away; the apostle reads the groaning as the lived face of a creation in productive travail under God. The two readings cannot be combined, because they differ at the level of what the groaning is *for*. A creation whose groans are the pains of childbirth is not the same kind of system as a creation whose groans are the warning signs of structural failure, and the response one owes it is not the same response.

The title of this paper — *The Entropic Cascade* — is owed to this passage. The cascade is not metaphor; it is the canonical category for what has happened to the world in the wake of the human act. And here is the point that returns this section to the paper's opening engagement with transhumanism: limitation and mortality are *symptoms* of the cascade. They are not the disease. Engineering symptoms while leaving the disease untreated is, at its most generous interpretation, a misdiagnosis; at its most candid, it is the serpent's logic at industrial scale. The transhumanist program offers to lift the futility from the creature by lifting the creature out of the creation. The Christian gospel offers to lift the futility from the creation by reconciling the creation to its Creator. The two offers are not competing strategies for the same end. They are competing ends.

4g. Entropy as Canonical Category: Physics, Biology, Mathematics

The title of this paper would be merely literary if the cascade Rom 8 names did not have technical referents in the sciences that describe the created order under their own categories. It does. The brief survey that follows is not a detour into natural philosophy for its own sake but a structural argument: that *mataiotēs* — the futility to which creation has been subjected — is the canonical theological name for a feature of the created order that physics, biology, and mathematics have, each in its own

²⁹David Wilkinson, *Christian Eschatology and the Physical Universe* (London: T&T Clark, 2010), doi:10.5040/9780567660497. The point relevant here is that the futility of Rom 8 is not a rewriting of natural law but a subjection of the created order to a corruption whose final reversal is the resurrection.

³⁰Klaus Nürnberger, "The 'Spirit' and the 'Groaning of Creation' in Romans 8," on the Spirit's role in the groaning of creation in Rom 8 and the pneumatological pledge of the future liberation.

register, independently identified. The convergence is not coincidence. Scripture is not borrowing a scientific metaphor; the sciences are noticing, under their own categories, a structural feature of the same created order Rom 8 names. Wilkinson's careful work on creation, cosmology, and new creation is the appropriate bridge for the argument, and the present section advances it by tracing the convergence one register at a time.³¹

Physics. The second law of thermodynamics, in the form Rudolf Clausius stated it in the mid-nineteenth century and Ludwig Boltzmann reframed statistically in its closing decades, holds that the entropy of an isolated system tends to increase. Useful energy degrades into less useful forms; ordered configurations of matter and energy tend, overwhelmingly, toward less ordered ones; the arrow of time itself, as the working physicist understands it, is the asymmetry the second law establishes. The cosmological extrapolation is the so-called heat death — the asymptotic condition of a universe in which all available energy has been redistributed to the point at which no further work is extractable from any gradient. The picture is one of irreversible degradation built into the operating principles of the material order. It is precisely *not* a picture of contingent decay produced by local accidents; it is the picture of a futility intrinsic to the running of the closed system. The convergence with Rom 8 is exact at the structural level: the apostle says the creation has been *subjected* to futility — the passive verb is load-bearing, the subjection is not the creature's choice — and that the subjection is *in hope* of a liberation that does not arise from within the subjected order. The thermodynamic picture, taken on its own terms, has no internal resource for hope; the subjected order it describes runs to its terminus without remedy. The theological picture, taken on its own terms, agrees about the subjection but locates the hope in what is given to the order from beyond it — the resurrection that introduces a new economy. The two pictures are not in tension. They are the same observation about the same created order made under two different lights.

Biology. Erwin Schrödinger's 1944 lectures, published as *What Is Life?*, offered the now-classic formulation that a living organism is a system that maintains itself far from thermodynamic equilibrium for a finite period by feeding on what he called *negative entropy* — the importation of order from the environment, exported as waste at higher entropy than it entered. Life, on this account, is local resistance to the universal thermodynamic trend, sustained for a time and finally lost. Death is the moment at which the negentropic resistance fails and the second law re-asserts itself locally. Aging is the cumulative degradation of the systems by which the resistance is mounted: the slow loss of the body's capacity to maintain its far-from-equilibrium

³¹Wilkinson, *Christian Eschatology and the Physical Universe*. The convergence-of-registers argument the present section develops is in the spirit of the monograph's project, which holds that the natural-scientific picture of the cosmos and the eschatological promises of the New Testament illuminate one another rather than standing in tension. The classic statements of the thermodynamic argument cited in the body are Rudolf Clausius's mid-nineteenth-century formulations of the second law and Ludwig Boltzmann's late-nineteenth-century statistical reframing. Erwin Schrödinger, *What Is Life? The Physical Aspect of the Living Cell* (Cambridge: Cambridge University Press, 1944), is the canonical statement of the negentropy argument for living systems. Claude E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal* 27 (1948): 379–423, 623–656, is the originating statement of the information-theoretic entropy measure. The relation between thermodynamic and information entropies, surfaced in Rolf Landauer's work on the thermodynamic cost of information erasure, is treated in the standard literature on the foundations of computing and physics.

configuration. None of this is in dispute as a description of what the biological order looks like when examined under its own categories. The theological reading does not contest the description. It contests the interpretation. The transhumanist program reads the negentropic resistance as a project to be extended indefinitely — to push the moment of failure further and further out and eventually past the visible horizon altogether. The Christian reading is different at the root: the resistance is finite *by design*, the design is not a defect, and the transformation that overcomes death is not the indefinite extension of the resistance but the introduction of a body, in resurrection, whose relation to entropy is not the present body's relation. The contrast matters. The transhumanist offers more time within the cascade. The Christian gospel offers a body outside it. The two are not competing strategies for the same outcome; they are different outcomes, and the difference is the difference between extending the present economy and entering a new one.

Mathematics. Claude Shannon's 1948 *Mathematical Theory of Communication* introduced an entropy measure — $H = -\sum p \log p$ — that quantifies the uncertainty in a probability distribution, or equivalently, the information content of a signal source. The mathematical and physical entropies are not the same quantity, but they are not unrelated either, as the subsequent literature on Landauer's principle has made clear: the erasure of information has a thermodynamic cost, and information theory and thermodynamics meet at the level of physical implementation. For the present argument, the relevant point is structural rather than technical. Communication channels degrade; signals are corrupted by noise; the shared codes by which messages are decoded across populations are subject to fragmentation. And here a striking exegetical observation is worth registering — held lightly, as illustrative analogy rather than as a claim that the text *is* an entropy event in the technical sense. The post-Fall narrative of Gen 11:1-9 records, as a judgment on the Babel project, *the multiplication of tongues*. The text labels this as the moment at which the human linguistic order is fragmented into incomprehensibility. In the mathematical idiom available since Shannon, the move the judgment performs is most precisely characterized not as an increase in the source entropy H of the species' utterances — which is a quantity internal to a code — but as a *collapse of channel capacity* and a loss of the *shared code* between groups: the mutual information that made cross-group decoding possible is destroyed, signals once carriers of meaning become noise to those outside the speaker's basin, and coordination at scale becomes intractable. The text does not, of course, use the modern category. But the modern category, when laid against the text, illuminates what the judgment is: the cascade reaches into the order of human communication itself, and the same structural shape — futility, the degradation of *usable* order — appears in the linguistic register as in the thermodynamic and the biological ones. The analogy is held as analogy. What it surfaces is that the order Rom 8 names futile is futile in registers the ancient text did not name and the modern sciences did not invent.

The convergence. Three independent scientific registers, developed without theological motivation and refined under their own internal disciplines, converge on a single observation: the created order, as it now stands, exhibits a structural tendency toward the degradation of usable order. Thermodynamic systems run toward heat death. Biological systems hold the trend off locally and finitely and then succumb. Communicative systems are subject to noise and to the fragmentation of the shared

codes on which decoding depends. The Christian doctrine of the cascade does not require the sciences in order to be true. But the sciences, taken on their own terms, supply an unsought corroboration of what the doctrine has named all along: that the order under which we live is one in which futility has been built into the operating principles, and that the human experience of decay, of mortality, and of the failure of our most ambitious coordinations is not a contingent misfortune but the lived face of a structural condition.

The convergence has two further consequences for the argument of this paper. First, it reframes the transhumanist diagnosis from the inside. The transhumanist is correct to notice that the present order is one in which limitation, decay, and death are universal and structural. The transhumanist is incorrect to infer that the appropriate response is the engineering of an exemption from the structure. The structure is not, on the Christian reading, the disease; the structure is the present face of a creation under judgment, and the remedy is not exemption but the new creation that Wilkinson's eschatological work — alongside Moo's exegesis of Rom 8 — develops as the proper Christian hope.³² Second, the convergence sharpens what is at stake in the federal headship of §4e. The cascade is not a localized human problem that local human interventions can address; it is the operating condition of the very order in which the interventions would be staged. Any response that does not reach the level of the order — that does not, in other words, introduce something into the order from beyond it — will be subject, by the order's own running, to the very condition it sought to escape. This is why the gospel cannot be merely a moral program and why the new creation cannot be merely an improved version of the old. The first paper of this series argued that the human is a particular kind of creature; the present paper argues that the creature is now in a particular kind of creation; the next paper will argue that the recovery is the work of one in whom both the creature and the creation are made new.

§5. What We Argue From

The warrant for the foregoing argument has been built across §4, but it is worth naming briefly. The argument is made from three sources, in the order of their authority. First, from Scripture: Gen 3 in its narrative force; Gen 9:6 on the persistence of the image after the Fall; Gen 11:1–9 on Babel and the inverted vocation; Rom 5:12–14 on the federal head and the cascade from one man; Rom 8:19–22 on the cosmic scope of the futility. The exegetical method is the redemptive-historical method recovered for the modern Reformed tradition by Geerhardus Vos and his successors, in which the text is read with attention to its place in the unfolding economy of the covenants.³³ Second, from the Reformed tradition: Calvin, Bavinck, Berkhof, Hoekema, Vos, Kline, Murray, Plantinga, Berkouwer — a representative rather than exhaustive list, each of whom carries part of the weight of the dogmatic argument at the points the exegesis requires it.³⁴ Third, from the analytical frame this series carries forward — what Pa-

³²Wilkinson, on new creation as the proper Christian hope set against the heat-death extrapolation; Moo, *Romans*, NICNT, *ad loc.* Rom 8:19–22, on the liberation of the creation as the structural answer to its subjection to futility.

³³Geerhardus Vos, *Biblical Theology: Old and New Testaments* (Grand Rapids: Eerdmans, 1948).

³⁴The representative citations are given through the body of the paper at the appropriate exegetical points. The list is meant to indicate the location of the present paper within a tradition, not to exhaust

per 1 introduced as HGC³AE², the lens through which the seven-problem set is loaded. The frame is named here rather than developed; it is the *lens* through which the series sees, not the *ground* on which the series argues. The ground is Scripture, read with the Reformed tradition. The lens is the series' analytic.

This brief naming is sufficient. The paper does not pause to defend its method at length, as that defense belongs to the series' opening matter and not to its second paper.

§6. If We Are Right

If the Fall is the real diagnosis — covenantal, anthropological, and cosmic — several consequences follow for the response the Christian tradition owes to the transhumanist program and to the wider register of self-deifying agendas of which transhumanism is the most explicit.

First, transhumanism is to be named, in its structure, as the serpent's offer in a new accent. This naming is not a slogan; it is the structural identification §4a developed. It must be done with honesty about the ache the program answers — the ache, that is, for the overcoming of death and limitation, an ache the Christian tradition does not invent, did not invent, and has its own account of. The fault of the program is not that it answers the ache. The fault is that the answer it gives is the answer the serpent already gave.

Second, the *imago* is the floor under every human regardless of capability. The categorical claim is Hoekema's and Bavinck's, but its practical import in the present moment is sharp.³⁵ Capability-as-worth is a category error, and the marred-but-real image refutes it. The un-augmented have the same worth as the augmented because the worth that matters is conferred and not achieved. The implications for any politics, any medicine, any economics that begins to drift toward the capability-as-worth premise are direct: the drift is, on Christian premises, idolatrous, because it locates the dignity of the person in a property that the person does not have in conferred relation to the Creator.

Third, work has a telos that *can* be redeemed but cannot be self-redeemed. The next paper in the series develops the redemption of vocation; the present paper insists only that Babel is the failure mode of any vocational program that attempts its own redemption. The cursed vocation is not self-uncursing. The work-making faculty is bent, and it is not the bent faculty that will straighten the work.

Fourth, the universal scope of the seven-problem set is covenantal, not contingent. This is the implication of Rom 5:12–14 carried into the practical question of where the response to the cascade must be lodged. Cultural reform alone does not address the cascade because the cascade is not, at its root, a cultural phenomenon. Reform is good in its place; the place of reform is not the place where the head is changed. Plantinga's prophetic catalogue of fallenness in *Not the Way It's Supposed to Be* is the helpful background here: the catalogue is honest because it refuses to settle for

the tradition's contributions on the doctrines treated.

³⁵Hoekema, *Created in God's Image*; Bavinck, *Reformed Dogmatics*, vol. 3.

any account of the wrong that does not name the root.³⁶

These four consequences are the practical fruit of the argument and the agenda the rest of the series takes up.

§7. The Protoevangelium: The Structural Pivot of the Series

At the very moment the seven problems are confirmed by the Fall, the first promise of the gospel is spoken inside the judgment. This is the structural pivot of the entire series, and the paper insists on it as such.

The pivot is exegetical and it is structurally located. “I will put enmity between you and the woman, and between your offspring and her offspring; he shall bruise your head, and you shall bruise his heel” (Gen 3:15, ESV). The verse is spoken to the serpent within the formal judgment scene of Gen 3:14–19. It is spoken *before* the expulsion from the Garden of Gen 3:23–24, and it is the only word of hope in the entire scene. The placement is load-bearing: the promise is not given after the cascade has played out and the human stands outside the Garden bereft; the promise is given *inside* the judgment, *before* the human leaves. The structural significance is that the cascade is, from its first verse, framed by promise. The seventh problem of the spine — purpose, end — is not abolished by the Fall. It is re-promised, in the seed of the woman, before the door of the Garden has even closed.

The exegetical literature here is load-bearing and primary, and the paper rests its case in §7 on two voices. R. A. Martin’s careful treatment of the earliest reception of the protoevangelium establishes that the messianic reading of Gen 3:15 is not a late Christian imposition on a more innocent text but is in fact the earliest reading the texts permit, traceable in the pre-Christian translation tradition.³⁷ Jacques Doukhan’s recent work, in turn, reads Gen 3:16 — the verse that immediately follows and is too often heard as unrelieved judgment — *in the light of* Gen 3:15. The result of that reading is not the softening of the curse but the recognition that the curse and the promise are spoken in one breath: the messianic hope is structurally present *inside* the judgment, not despite it.³⁸ Doukhan’s contribution preserves the seriousness of the judgment while refusing to detach the judgment from its immediately surrounding promise. This is the careful work the structural-pivot reading requires.

Vos’s redemptive-historical method is the appropriate frame for receiving the pivot.³⁹ The protoevangelium is not, on this reading, a stray theological flourish embedded in a pre-theological narrative. It is the inaugural moment of the redemptive history the rest of Scripture unfolds — and it is positioned, by the divine speaker, precisely at the moment the cascade begins, so that the cascade and the promise are inseparable from one another in the canonical structure. The seed of the woman who will bruise the

³⁶Plantinga, *Not the Way It’s Supposed to Be*.

³⁷R. A. Martin, “The Earliest Messianic Interpretation of Genesis 3:15,” *Journal of Biblical Literature* 84, no. 4 (1965), on the pre-Christian reception of the verse.

³⁸Jacques B. Doukhan, “Tragic Curse, or Messianic Hope? Reading Genesis 3:16 in Light of Gen 3:15,” *Religions* 15 (2024), preserving the seriousness of the judgment while refusing to detach it from its immediately surrounding promise.

³⁹Vos, *Biblical Theology*. The redemptive-historical method, applied to the protoevangelium, requires precisely the structural-pivot reading the present section develops.

serpent's head is the figure whose obedience Rom 5:15–21 will trace as the recovery of the federal headship lost in Gen 3, and that recovery is the burden of the next paper in this series, not of the present one. The present paper insists only on the pivot: Problem #7 is *advanced*, not *closed*, by Gen 3:15, and the trajectory the series will trace is set by this verse.

The cascade is entropic. It is not terminal. The reason it is not terminal is not that the cascade is shallow but that the promise is present at its inception. The image is defaced but not erased; the curse is heavy but not the last word spoken in the judgment scene; the federal head is fallen but the second Adam is already promised in seed.

A structural analog from the mathematics of dynamical systems can clarify, held lightly and at proper subordination, what the canonical pivot accomplishes. In non-linear systems the *bifurcation point* is a value of a parameter at which the qualitative behavior of the system's trajectories changes — at which the long-run *basin of attraction* governing where the trajectories terminate is reorganized — and the famous insight, sharpened in Edward Lorenz's mid-twentieth-century atmospheric work and developed in the wider chaos- and dynamical-systems literature that followed, is that the qualitative destination of the system can be set by what occurs at the bifurcation in a way that no subsequent intervention along either trajectory can simply undo. The analog must be offered with care and worn lightly: the providence of God is not a deterministic dynamical system, the language of basins is not the language of grace, and the canonical trajectory is governed by personal divine speech and ongoing divine action rather than by parameter values run forward by the system itself. What the analog surfaces is the structural feature on which the theological claim trades — that Gen 3:15 is spoken at the precise moment the seven-problem set is confirmed and the curse is being articulated, *inside* the judgment, *before* the expulsion, *before* any human action under the new conditions has occurred — and that the verse functions, in the canonical economy, as the divinely uttered word at which the qualitative destination of the post-Fall human race is set. The cascade continues; its terminus is no longer the cascade.

What the redemptive-historical method recovers when it reads Gen 3:15 as the pivot is precisely this load-bearing structural fact: that the post-Fall economy of Scripture is dual-trajectoried by divine speech from its outset, and that the basin into which the redeemed creation will settle is the basin the pivot opened.⁴⁰ The history that follows from Gen 3:16 onward — the expulsion, Cain, the antediluvian violence, the flood, Babel, the call of Abram, the patriarchs, the exodus, the law, the prophets, and the long preparation for the coming of the one in whom the promise is fulfilled — is the trajectory the pivot has set. Doukhan's reading of Gen 3:16 *in the light of* Gen 3:15 is the exegetical instrument that surfaces what is structurally at work in the canon: the curse and the promise are spoken in one breath, and the curse is heard rightly only when the promise is heard alongside it. The cascade is entropic; the cascade is not terminal. It is not terminal because the seed of the woman is promised at its inception, and because the work of subsequent papers in this series is the tracing of the redemptive trajectory that promise opened — toward the second Adam, toward

⁴⁰Vos, *Biblical Theology*. The redemptive-historical method, applied to the protoevangelium, requires precisely the structural-pivot reading the present section develops.

the new creation, and toward the consummation that Rom 8 holds in hope.

§8. Conclusion

Paper 1 grounded a seven-problem set in the doctrine of creation. The present paper has argued that the Fall confirms that set — intensifies it, universalizes it through the covenantal cascade of Rom 5:12-14, and extends it into the cosmic order of Rom 8:19-22. The serpent’s offer in Gen 3:5 is the structural ancestor of the modern transhumanist program; that program is not a strawman but the most candid contemporary statement of *eritis sicut Deus* in technological register. Against that offer, the paper has insisted that the image of God is defaced but not erased, that vocation is cursed but not abolished, that governance is distorted but not retired, and that the federal head’s fall is not the last word because the seed of the woman is promised at the very moment of the curse.

The cascade is entropic. The cascade is not terminal. The structural pivot of the series is Gen 3:15. The work of the next paper is the recovery the second Adam accomplishes — the “much more” of Rom 5:15-21, the obedience by which the *many* are made righteous, and the trajectory by which the seven problems, confirmed here in their fallen register, are addressed at the root rather than at the symptom. The present paper terminates with the pivot in view, having done the diagnostic work the rest of the series presupposes.

Footnotes